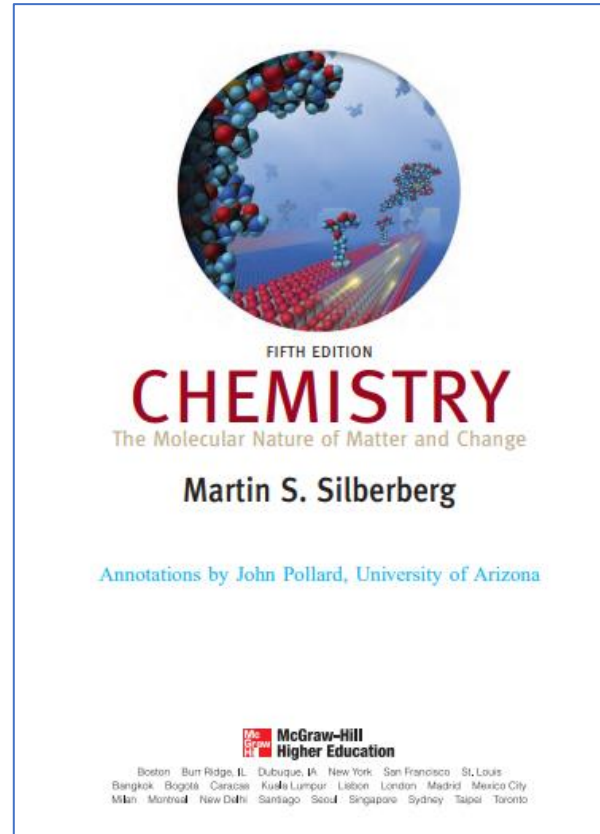


GENERAL CHEMISTRY – CHE 1201

Chemistry of Elements

P-block Carbon family

Referencing Text



14.6 GROUP 4A(14): THE CARBON FAMILY

KEY

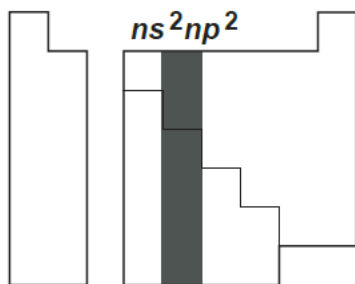
Atomic No.

Symbol

Atomic mass

Valence e⁻
configuration

Common
oxidation
states



GROUP 4A(14)

6 C 12.01 $2s^22p^2$ (-4, +4, +2)	
14 Si 28.09 $3s^23p^2$ (-4, +4)	
32 Ge 72.61 $4s^24p^2$ (+4, +2)	
50 Sn 118.7 $5s^25p^2$ (+4, +2)	
82 Pb 207.2 $6s^26p^2$ (+4, +2)	
114 (285) $7s^27p^2$	Observed in experiments at Dubna, Russia, in 1998

Table 14.2 Bond Type and the Melting Process in Groups 3A(13) to 5A(15)

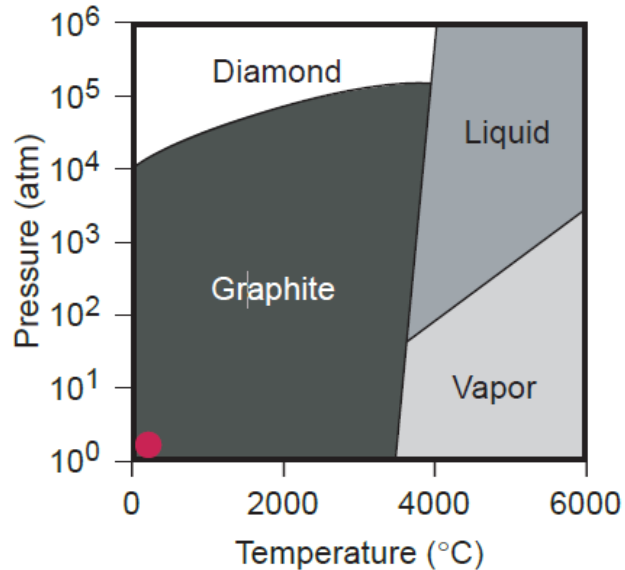
Period	Group 3A(13)				Group 4A(14)				Group 5A(15)				Key:
	Element	Bond Type	Melting Point (°C)	ΔH_{fus} (kJ/mol)	Element	Bond Type	Melting Point (°C)	ΔH_{fus} (kJ/mol)	Element	Bond Type	Melting Point (°C)	ΔH_{fus} (kJ/mol)	
2	B		2180	23.6	C		4100	Very high	N		-210	0.7	Metallic
3	Al		660	10.5	Si		1420	50.6	P		44.1	2.5	Covalent network
4	Ga		30	5.6	Ge		945	36.8	As		816	27.7	Covalent molecule
5	In		157	3.3	Sn		232	7.1	Sb		631	20.0	Metal
6	Tl		304	4.3	Pb		327	4.8	Bi		271	10.5	Metalloid
													Nonmetal

- Within Group 4A, the large decrease in melting point between the network covalent solids C and Si is due to longer, weaker bonds in the Si structure and the large decrease between Ge and Sn is due to the change from covalent network to metallic bonding.
- Considering horizontal trends, the large increases in melting point and ΔH_{fus} across a period between Al and Si and between Ga and Ge reflect the change from metallic to covalent network bonding.
- Note the unexpected rises in the values for these properties from metallic Al, Ga, and Sn to the network covalent metalloids Si, Ge, and Sb, and note the abrupt drops from the covalent networks of C and Si to the individual molecules of N and P in Group 5A.

Allotropism: Different Forms of an Element

Allotropes: different crystalline or molecular forms of a substance.

One allotrope is usually more stable than another at a particular pressure and temperature.



- Graphite is a black electrical conductor that is soft and “greasy,” whereas diamond is a colorless electrical insulator that is extremely hard.
- Graphite is the standard state of carbon, the more stable form at ordinary temperature and pressure.
- Fortunately for jewelry owners, diamond changes to graphite at a negligible rate under normal conditions.

Figure 14.12 Phase diagram of carbon. Graphite is the more stable form of carbon at ordinary conditions (*small red circle at extreme lower left*). Diamond is more stable at very high pressure.

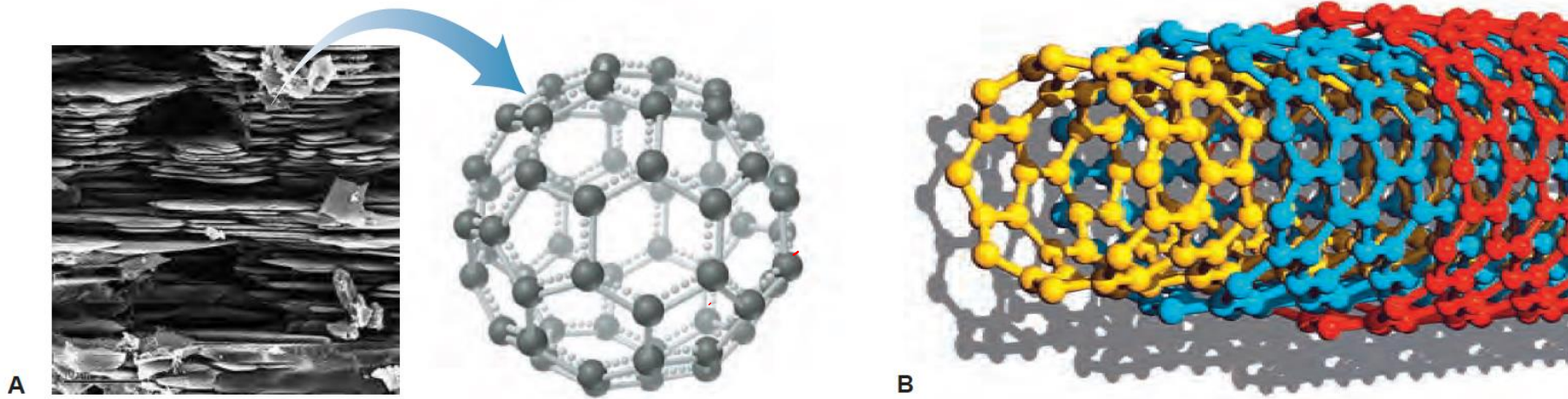


Figure 14.13 Buckyballs and nanotubes. **A**, Crystals of buckminsterfullerene (C₆₀) next to a ball-and-stick model. The parent of the fullerenes, the “buckyball,” is a soccer ball-shaped molecule of 60 carbon atoms. **B**, Nanotubes are single or, as shown in this colored model, concentric graphite-like tubes (shown here without the usual fullerene ends; see the Gallery, p. 398).

Tin (Sn) has two allotropes

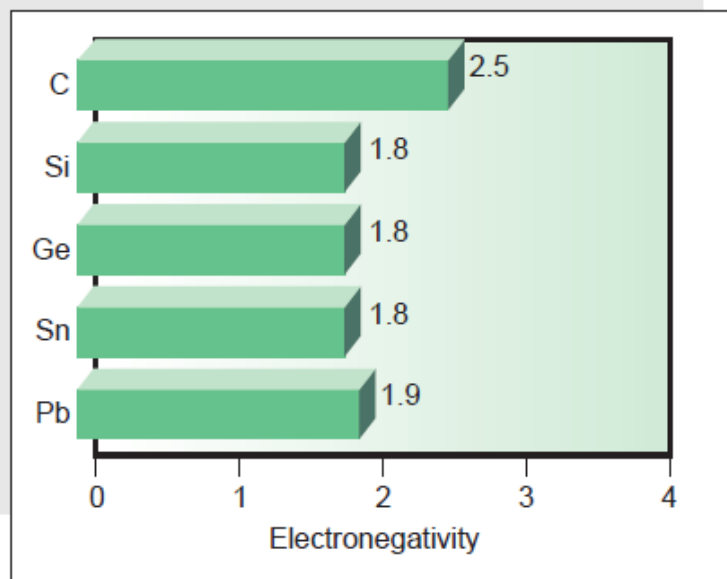
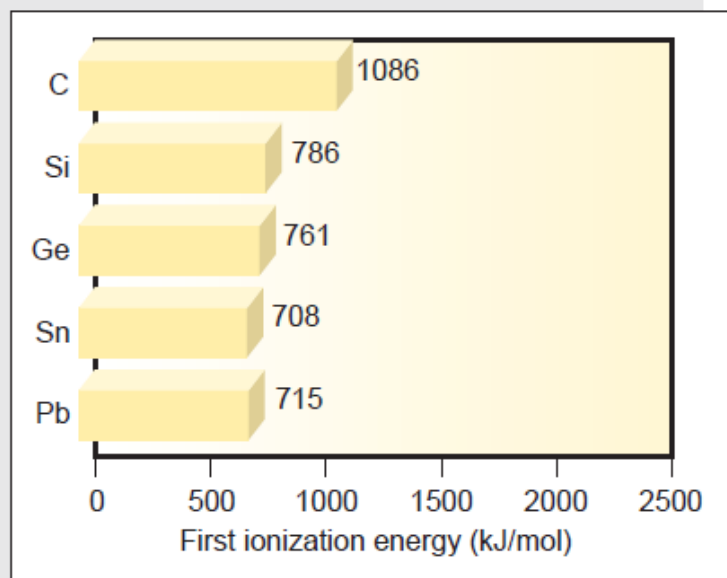
- White -tin is stable at room temperature and above, whereas gray -tin is the more stable form below 13°C.
- When white tin is kept for long periods at a low temperature, some converts to microcrystals of gray tin.
- The random formation and growth of these regions of gray tin, which has a different crystal structure, weaken the metal and make it crumble.

Atomic Properties

Atomic radius (pm)		Ionic radius (pm)
C 77		
Si 118		
Ge 122		
Sn 140		Sn ²⁺ 118
Pb 146		Pb ²⁺ 119

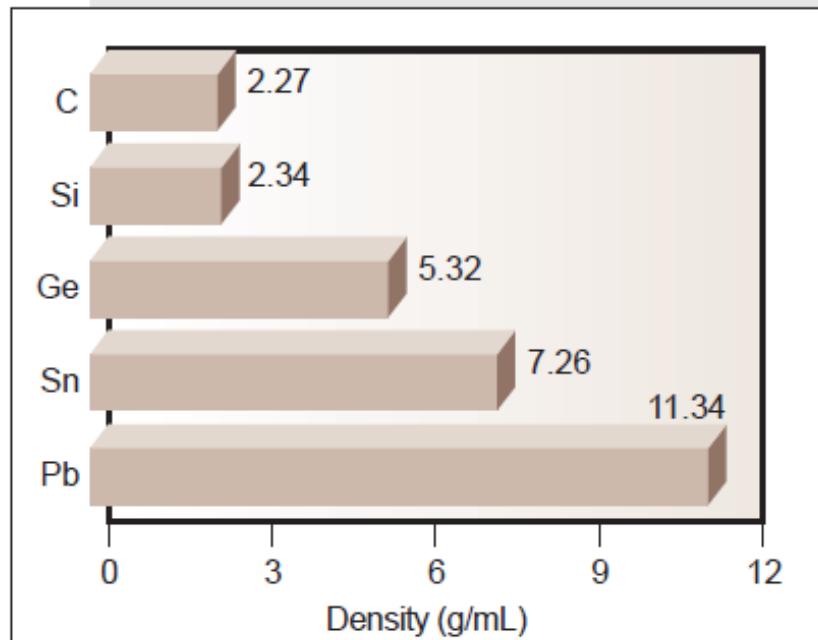
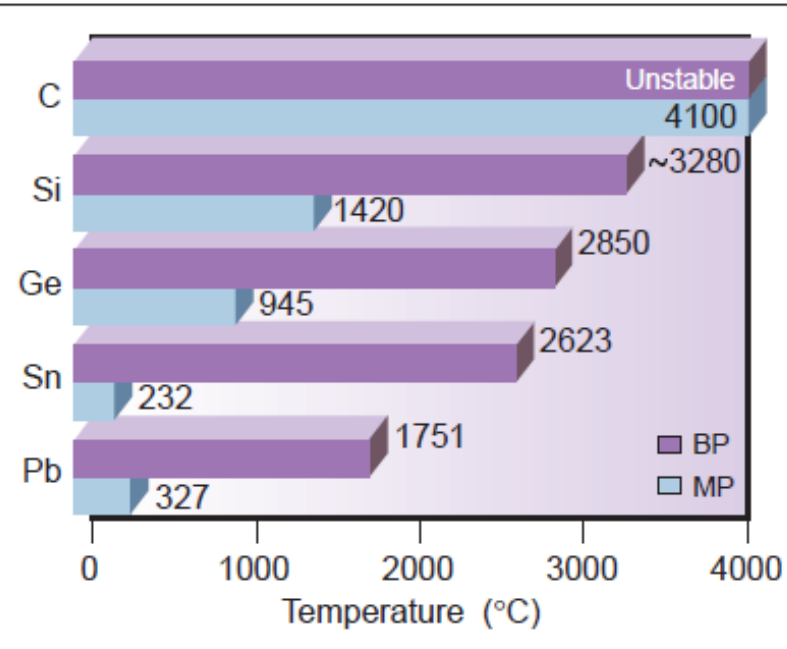
Group electron configuration is ns^2np^2 . Down the group, the number of oxidation states decreases, and the lower (+2) state becomes more common.

Down the group, size increases. Because transition and inner transition elements intervene, IE and EN do not decrease smoothly.



Physical Properties

Trends in properties, such as decreasing hardness and melting point, are due to changes in types of bonding within the solid: network covalent in C, Si, and Ge; metallic in Sn and Pb (*see text*).



Down the group, density increases because of several factors, including differences in crystal packing.

Some Reactions and Compounds

Important Reactions

Reactions 1 and 2 pertain to all the elements (E); reactions 3 to 7 concern industrial uses for compounds of C and Si.

1. The elements are oxidized by halogens:

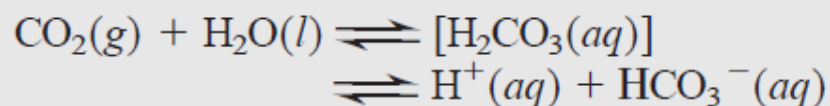


The +2 halides are more stable for tin and lead, SnX_2 and PbX_2 .

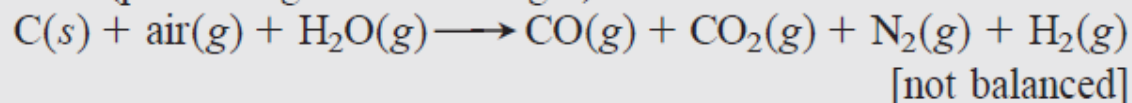
2. The elements are oxidized by O_2 :



Pb forms the +2 oxide, PbO . Oxides become more basic down the group. The reaction of CO_2 and H_2O provides the weak acidity of natural unpolluted waters:



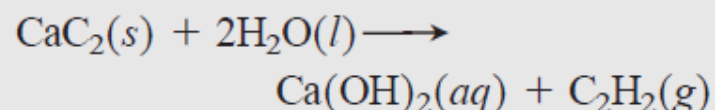
3. Air and steam passed over hot coke produce gaseous fuel mixtures (producer gas and water gas):



4. Hydrocarbons react with O_2 to form CO_2 and H_2O . The reaction for methane is adapted to yield heat or electricity:

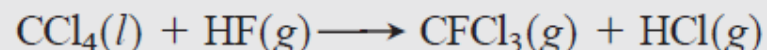


5. Certain metal carbides react with water to produce acetylene:



The gas is used to make other organic compounds and as a fuel in welding (*see photo*).

6. Freons (chlorofluorocarbons) are formed by fluorinating carbon tetrachloride:



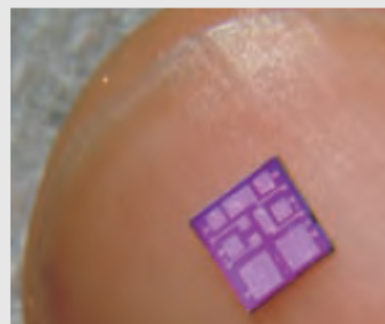
Production of trichlorofluoromethane (Freon-11), the most widely used refrigerant in the world, is being eliminated because of adverse effects on the environment (*margin note, p. 589*).

7. Silica is reduced to form elemental silicon:



This crude silicon is made ultrapure through zone refining (*Gallery, p. 533*) for the manufacture of computer chips (*see photo*).

Oxyacetylene torch



Computer chip

Important Compounds

1. **Carbon monoxide, CO.** used as a gaseous fuel, as a precursor for organic compounds, and as a reactant in the purification of nickel. Formed in internal combustion engines and released as a toxic air pollutant.
2. **Carbon dioxide, CO₂.** Atmospheric component used by photosynthetic plants to make carbohydrates and O₂. The final oxidation product of all carbon-based fuels; its increase in the atmosphere is changing Earth's climate. Used industrially as a refrigerant gas, a blanketing gas in fire extinguishers, and an effervescent gas in beverages. Combined with NH₃ to form urea for fertilizers and plastics manufacture.
3. **Methane, CH₄.** Used as a fuel and in the production of many organic compounds. Major component of natural gas. Formed by anaerobic decomposition of plants (swamp gas) and by microbes in termites and certain mammals. Greenhouse gas that contributes to global warming.

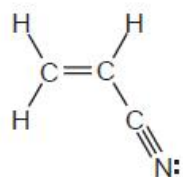
Important Compounds

4. **Silicon dioxide, SiO_2** . Occurs in many amorphous (glassy) and crystalline forms, quartz being the most common. Used to make glass and as an inert chromatography support material.
5. **Silicon carbide, SiC** . Known as carborundum, a major industrial abrasive and a highly refractory ceramic for tough, high temperature uses. Can be doped to form a high-temperature semiconductor.
6. **Organotin compounds, R_4Sn** . Used to stabilize PVC [poly(vinyl chloride)] plastics and to cure silicone rubbers.

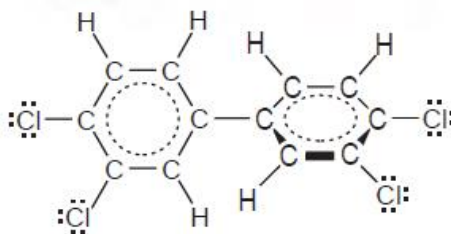
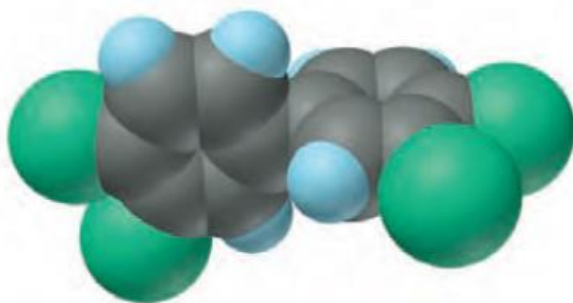
Agricultural biocide for insects, fungi, and weeds.
7. **Tetraethyl lead, $(\text{C}_2\text{H}_5)_4\text{Pb}$** . Once used as a gasoline additive to improve fuel efficiency, but now banned because of its inactivation of auto catalytic converters. Major source of lead as a toxic air pollutant.

Highlights of Carbon Chemistry

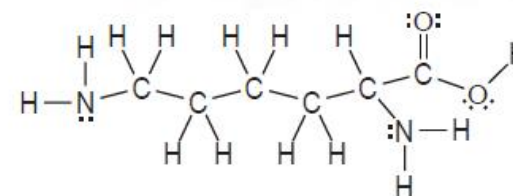
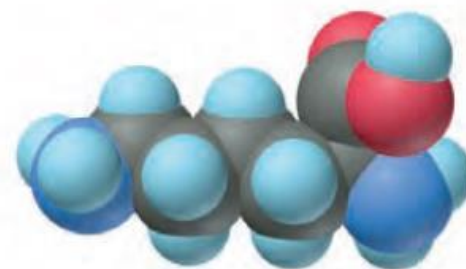
- Carbon forms bonds with the smaller Group 1A(1) and 2A(2) metals, many transition metals, the halogens, and its neighbors B, Si, N, O, P, and S.
- It also exhibits every oxidation state possible for its group, from +4 in CO_2 and halides like CCl_4 through -4 in CH_4 .
- Two features stand out in the chemistry of carbon: ability to bond to itself to form chains and to form multiple bonds.
- As a result of its small size and its capacity for four bonds, carbon can form chains, branches, and rings that lead to myriad structures. Add a lot of H, some O and N, a bit of S, P, halogens, and a few metals, and you have the whole organic world!



Acrylonitrile



PCB



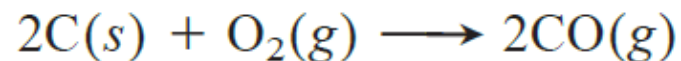
Lysine

Highlights of Carbon Chemistry

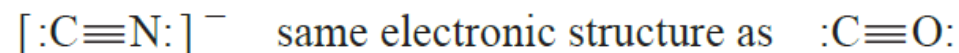
- Multiple bonds are common in carbon structures because the C-C bond is short enough for side-to-side overlap of two half-filled 2p orbitals to form π bonds.
- Because the other 4A members are larger, E-E bonds become longer and weaker down the group; thus, in terms of bond strength, C-C > Si-Si > Ge-Ge. The empty d orbitals of these larger atoms make the chains much more susceptible to chemical attack.
- Thus, some compounds with long Si chains are known but they are very reactive, and the longest Ge chain has only eight atoms.
- Moreover, longer bonds do not typically allow sufficient overlap of p orbitals for π bonding. Only very recently have compounds with π bonds between Si atoms or between Ge atoms been prepared, and they are also extremely reactive.
- In contrast to its organic compounds, carbon's inorganic compounds are simple. Metal carbonates are the main mineral form. Marble, limestone, chalk, coral, and several other types are found in enormous deposits throughout the world.
- Many of these compounds are leftovers of fossilized marine organisms. Carbonates are used in several common antacids because they react with the HCl in stomach acid



- Unlike the other 4A members, which form only solid network-covalent or ionic oxides, carbon forms two common gaseous oxides, CO₂ and CO.
- Carbon dioxide is essential to all life: it is the primary source of carbon in plants and animals through photosynthesis.
- Its aqueous solution is the cause of acidity in natural waters.
- However, its atmospheric buildup from deforestation and excessive use of fossil fuels is severely affecting the global climate.
- Carbon monoxide forms when carbon or its compounds burn in an inadequate supply of O₂

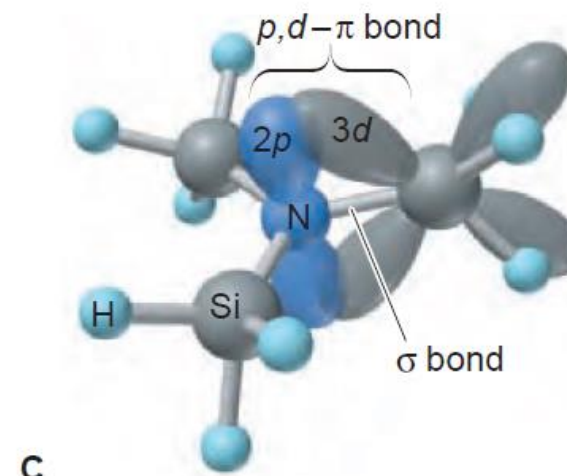
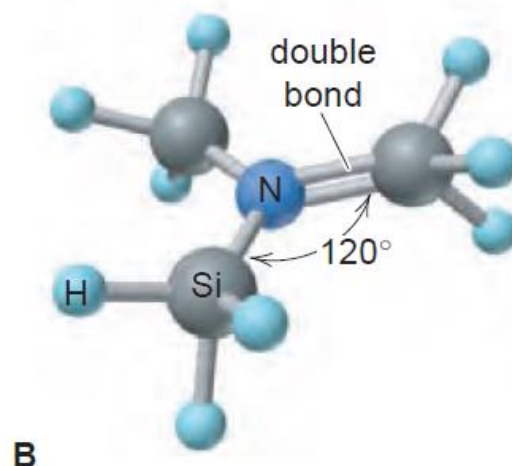
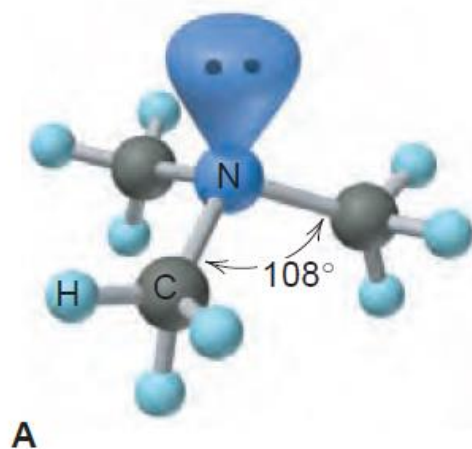


- Carbon monoxide is a key component of syngas fuels and is widely used in the production of methanol, formaldehyde, and other major industrial compounds.
- CO binds strongly to many transition metals. When inhaled in cigarette smoke or polluted air, it enters the blood and binds strongly to the Fe(II) in hemoglobin, preventing the normal binding of O₂, and to other iron-containing proteins.
- The cyanide ion (CN⁻) is isoelectronic with CO:



Highlights of Silicon Chemistry

- Silicon halides are more reactive than carbon halides because Si has empty d orbitals that are available for bond formation.
- One explanation for this unusual strength is that the Si-X bond has some double-bond character because of the presence of a σ bond and a different type of π bond, called a $p_\pi - d_\pi$ bond. This bond arises from side-to-side overlap of an Si d orbital and a halogen p orbital.
- A similar type of $p, d-\pi$ bonding leads to the trigonal planar molecular shape of trisilylamine, in contrast to the trigonal pyramidal shape of trimethylamine.



- Just as carbon forms unending C-C chains, the Si-O grouping repeats itself endlessly in a wide variety of **silicates**, the most important minerals on the planet, and in **silicones**, synthetic polymers that have many applications:

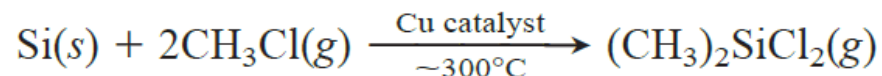
1. Silicate minerals.

- Oxygen, the most abundant element on Earth, and silicon, the next most abundant, compose these minerals and thus account for four of every five atoms on the surface of the planet!



2. Silicone polymers

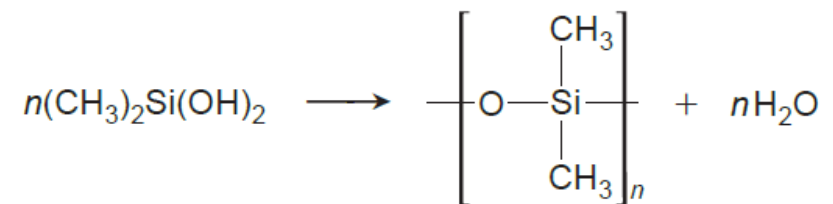
- A key starting material is formed by the reaction of silicon with methyl chloride:



Although the corresponding carbon compound $[(\text{CH}_3)_2\text{CCl}_2]$ is inert in water, the empty d orbitals of Si make $(\text{CH}_3)_2\text{SiCl}_2$ reactive:



The product reacts with itself to form water and a silicone chain molecule called a *poly(dimethyl siloxane)*:



- Silicones have properties of both plastics and minerals. The organic groups give them the flexibility and weak intermolecular forces between chains that are characteristic of a plastic, while the O-Si-O backbone confers the thermal stability and nonflammability of a mineral.

Diagonal Relationships: Boron and Silicon

metalloids boron and silicon.

- Both of these elements exhibit the electrical properties of a semiconductor
- Both B and Si and their mineral oxoanions (i.e; borates and silicates) occur in extended covalent networks.
- Both boric acid $[B(OH)_3]$ and silicic acid $[Si(OH)_4]$ are weakly acidic solids that held together by H bonding.
- Their hydrides (boranes silanes) are both flammable, low-melting compounds that act as reducing agents.